# LECTURE 11
# Basics of Discretisation Methods

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapter 3, pp. 45–78.

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# How do we actually solve these equations?

General PDE:

$$\boxed{\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \nabla \cdot \mathbf{F}_d(\mathbf{U}, \nabla \mathbf{U}) + \mathbf{S}}$$

**LHS:**

Storage in time plus the divergence of a flux — the conservative core.

**RHS:**

a diffusive flux depending on gradients, and a source term.

**Euler:** no diffusive flux, no source terms.

**Navier–Stokes**: both. The structure never changes.

# Important questions:

**1. Space:**

How do I represent $\partial/\partial x$ on a finite set of points?

**2. Time:**

How do I advance the solution from one level to the next?

**3. Boundaries:**

How do I impose conditions on the edge of the mesh?

**4. Algebra:**

How do I solve the system of algebraic equations I produced?

# Our solution scheme:

**Build the tools:** Taylor series into difference formulas -> the difference operators

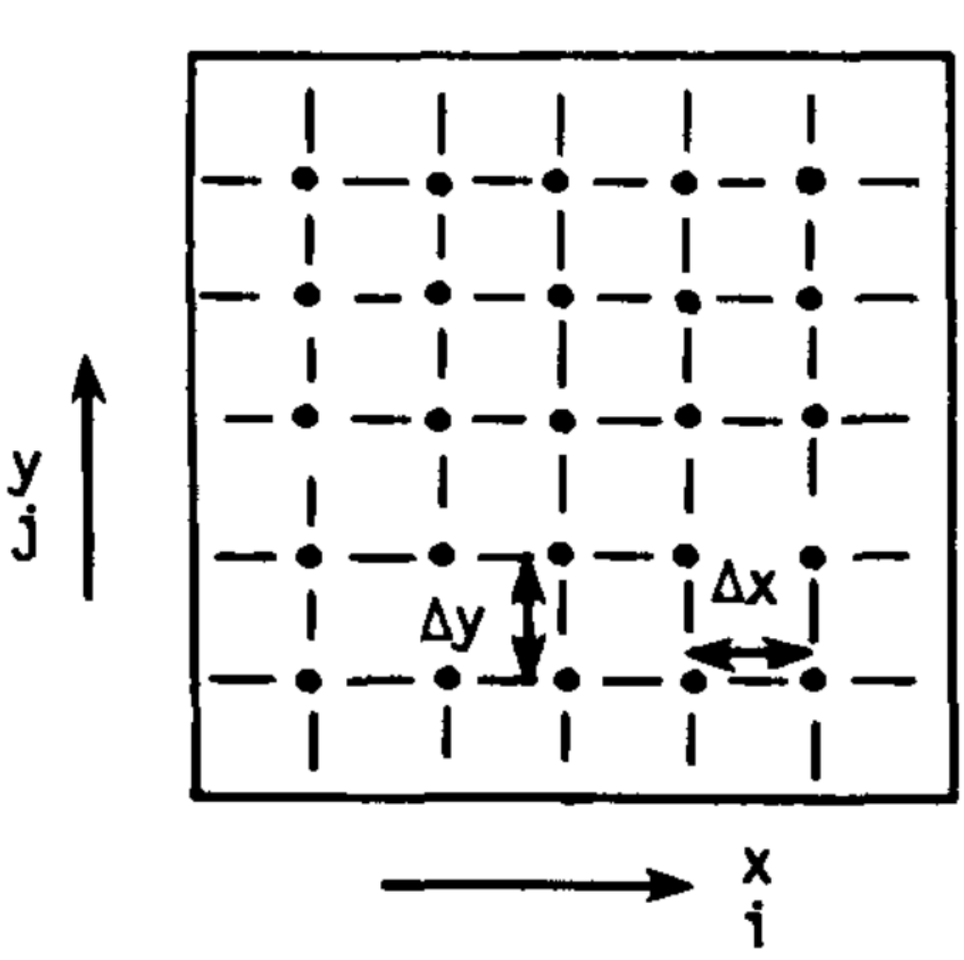
**Apply them to a PDE:** the heat equation, differenced four different ways.

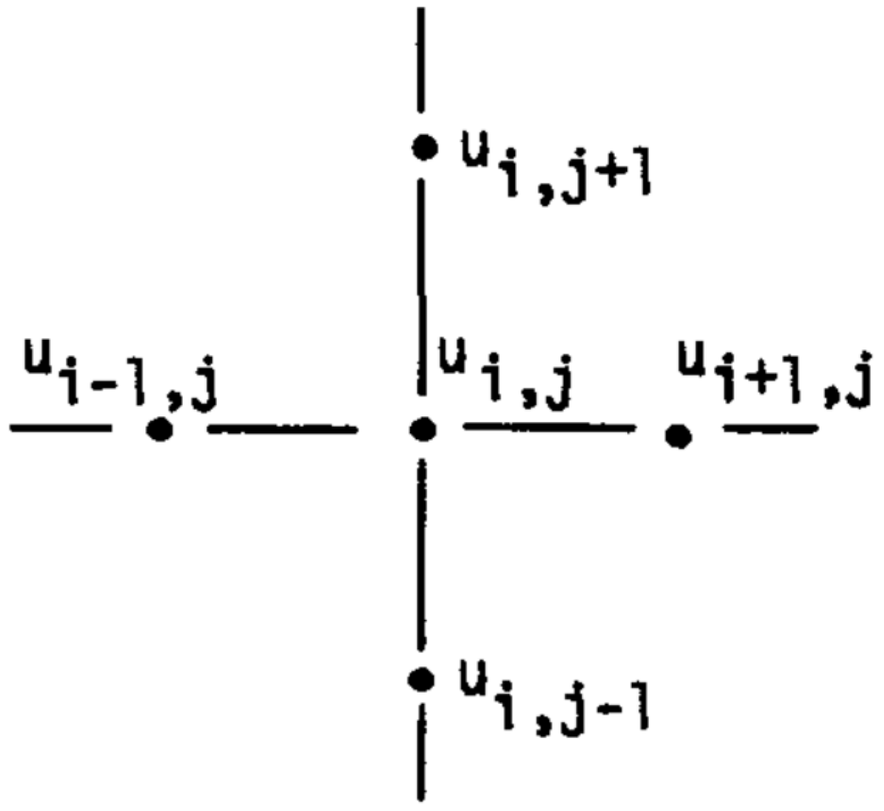
**Judge the result:** truncation error, consistency, stability, convergence.

**Get the physics right:** conservation form, and the conservative property of a scheme.

# 1. Discretising the domain

Replace the continuous domain by a mesh.
The dependent variable now exists only at the grid points.

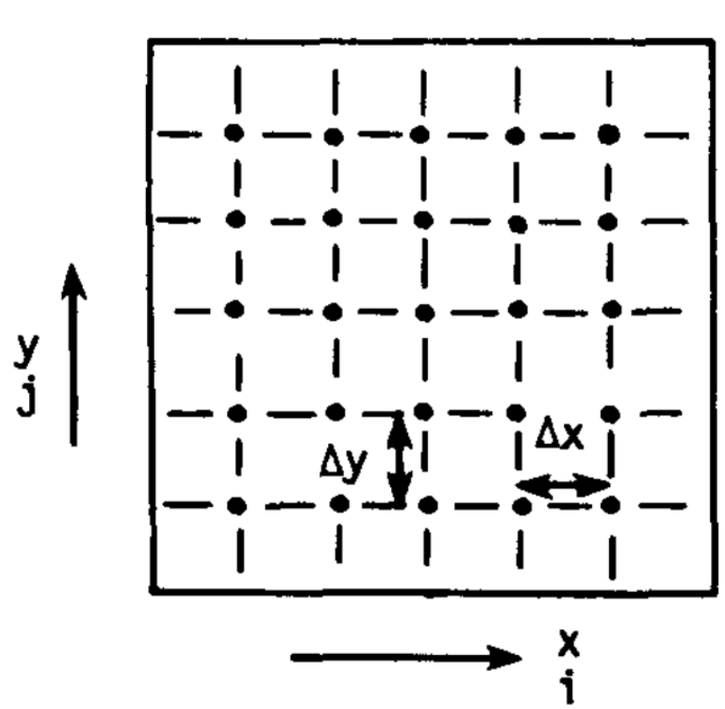


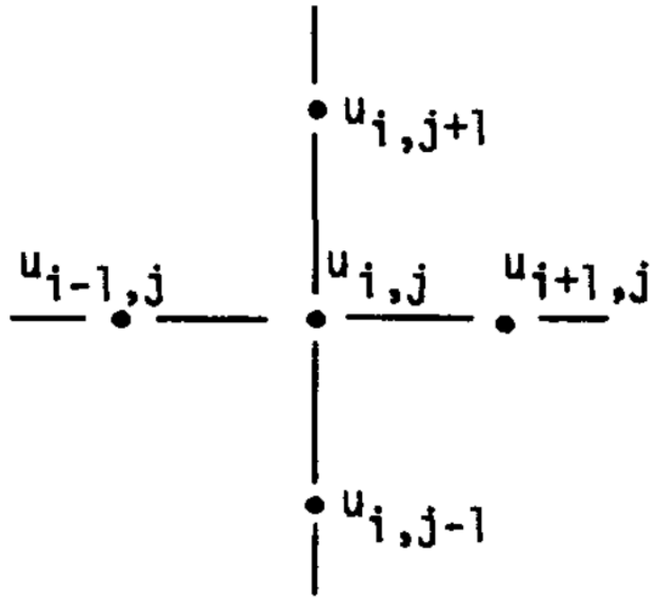




*A typical finite-difference grid, and the labelling of a point and its neighbours (Tannehill).*

$$x \to i\,, \quad y \to j\,, \quad z \to k\,, \quad t \to n \qquad \Longrightarrow \qquad u(x,y,z,t) \longrightarrow u^n_{i,j,k}$$

# Neighbours of the point $(i, j)$




$$u_{i+1,j} = u(x_0 + \Delta x,\, y_0)\,, \qquad u_{i-1,j} = u(x_0 - \Delta x,\, y_0)$$

$$u_{i,j+1} = u(x_0,\, y_0 + \Delta y)\,, \qquad u_{i,j-1} = u(x_0,\, y_0 - \Delta y)$$

Difference equations are written in terms of the general point and its neighbours, so one formula serves every interior point.

Notice that 1 algebraic relation, applied at every point of the mesh, replaces the PDE.

# From a derivative to a finite difference equation (FDE)

We start from the definition of the derivative without taking the limit:

$$\left.\frac{\partial u}{\partial x}\right|_{x_0,y_0} = \lim_{\Delta x \to 0} \frac{u(x_0 + \Delta x,\, y_0) - u(x_0,\, y_0)}{\Delta x}$$

For a small but finite $\Delta x$ the quotient is an approximation.

The **mean-value theorem** says it is exact somewhere in the interval.

This is an **approximation** to the derivative, so important questions are:

- How wrong is it?
- How fast does that improve?

# Taylor series: the forward difference

$$u(x_0 + \Delta x,\, y_0) = u(x_0, y_0) + \left.\frac{\partial u}{\partial x}\right|_0 \Delta x + \left.\frac{\partial^2 u}{\partial x^2}\right|_0 \frac{(\Delta x)^2}{2!} + \left.\frac{\partial^3 u}{\partial x^3}\right|_0 \frac{(\Delta x)^3}{3!} + \cdots$$

We start from the Taylor expansion above, and then solve for the first derivative.

What is left over is the **truncation error:**

$$\left.\frac{\partial u}{\partial x}\right|_{i,j} = \underbrace{\frac{u_{i+1,j} - u_{i,j}}{\Delta x}}_{\text{difference}} \underbrace{- \left.\frac{\partial^2 u}{\partial x^2}\right|_0 \frac{\Delta x}{2!} - \cdots}_{\text{truncation error}} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + O(\Delta x)$$

Therefore, **Truncation Error = PDE − FDE**
Here the leading term is proportional to $\Delta x$, so the scheme is first-order accurate.

$$\text{T.E.} = O(\Delta x) \quad \Longleftrightarrow \quad |\text{T.E.}| \leq K\,|\Delta x| \quad \text{as} \;\; \Delta x \to 0$$

$$K > 0 \;\; \text{constant, independent of } \Delta x$$

# What is the order of accuracy, $O(\Delta x)$?

$O(\Delta x)$ indicates how the error behaves as the mesh is refined: it is a rate, not a magnitude.

It does not say that an $O[(\Delta x)^2]$ scheme is better than an $O(\Delta x)$ scheme necessarily.

The order is the smallest power of $\Delta x$ common to every term of the truncation error.

# Backward and central differences

We can expand backwards instead:

$$u(x_0 - \Delta x,\, y_0) = u(x_0, y_0) - \left.\frac{\partial u}{\partial x}\right|_0 \Delta x + \left.\frac{\partial^2 u}{\partial x^2}\right|_0 \frac{(\Delta x)^2}{2!} - \left.\frac{\partial^3 u}{\partial x^3}\right|_0 \frac{(\Delta x)^3}{3!} + \cdots$$

Subtracting the 2 expansions removes the 2nd-derivative term and gives the central difference:

$$\left.\frac{\partial u}{\partial x}\right|_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2\,\Delta x} + O\left[(\Delta x)^2\right]$$

It considers the same 2 neighbours and same cost with better order of accuracy.

# Diffusion term: the second derivative

Adding the 2 expansions instead of subtracting removes the first derivative:

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{(\Delta x)^2} + O\left[(\Delta x)^2\right]$$

Notice the properties:

- We have 3 points,
- it is second-order accurate, and
- symmetric.

This is the **stencil** behind every diffusion term that appears in code.

**We read it as a curvature:**

The value at the centre compared with the average of its 2 neighbours.

# Finite-difference approximations

| **Scheme** | **Approximation to** $\partial u/\partial x\|_{i,j}$ | **T.E.** |
|---|---|---|
| Forward | $\dfrac{u_{i+1,j} - u_{i,j}}{h}$ | $O(h)$ |
| Backward | $\dfrac{u_{i,j} - u_{i-1,j}}{h}$ | $O(h)$ |
| Central | $\dfrac{u_{i+1,j} - u_{i-1,j}}{2h}$ | $O(h^2)$ |
| One-sided, 3 pt | $\dfrac{-3u_{i,j} + 4u_{i+1,j} - u_{i+2,j}}{2h}$ | $O(h^2)$ |

Notice that second-order one-sided formulas exist and are important.

For example, at a boundary you cannot reach across, but you still want $O(h^2)$.

# Difference operators: Tannehill notation

Forward Δ, backward ∇, central $\bar{\delta}$ and $\delta$, averaging $\mu$, shift $E$:

$$\Delta_x u_{i,j} = u_{i+1,j} - u_{i,j} \quad (3.9)$$

$$\nabla_x u_{i,j} = u_{i,j} - u_{i-1,j} \quad (3.11)$$

$$\bar{\delta}_x u_{i,j} = u_{i+1,j} - u_{i-1,j} \quad (3.13)$$

$$\delta_x u_{i,j} = u_{i+1/2,j} - u_{i-1/2,j} \quad (3.14)$$

$$\mu_x u_{i,j} = \tfrac{1}{2}\left(u_{i+1/2,j} + u_{i-1/2,j}\right) \quad (3.16)$$

$$I\, u_{i,j} = u_{i,j}\,, \qquad E_x^s u_{i,j} = u_{i+s,j}$$

$$\delta_x^2 u_{i,j} = \delta_x(\delta_x u_{i,j}) = u_{i+1,j} - 2u_{i,j} + u_{i-1,j} \quad (3.15)$$

$\bar{\delta}_x$ spans a full cell, $\delta_x$ half a cell.
Only $\delta_x$ composes, so $\delta_x^2$ is a 3-point stencil.

$$\left.\frac{\partial u}{\partial x}\right|_{i,j} = \frac{\Delta_x u_{i,j}}{\Delta x} + O(\Delta x) \quad (3.10)$$

$$\left.\frac{\partial u}{\partial x}\right|_{i,j} = \frac{\bar{\delta}_x u_{i,j}}{2\,\Delta x} + O[(\Delta x)^2] \quad (3.19)$$

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{i,j} = \frac{\delta_x^2 u_{i,j}}{(\Delta x)^2} + O[(\Delta x)^2] \quad (3.20)$$

# Mixed derivatives

For a mixed derivative we have one difference applied to another.
Taking both central gives:

$$\left.\frac{\partial^2 u}{\partial x \partial y}\right|_{i,j} = \frac{1}{2\,\Delta x}\left(\frac{u_{i+1,j+1} - u_{i+1,j-1}}{2\,\Delta y} - \frac{u_{i-1,j+1} - u_{i-1,j-1}}{2\,\Delta y}\right)$$

$$+ O\left[(\Delta x)^2, (\Delta y)^2\right]$$

Tannehill's Table 3.2 lists a dozen variants.

They differ in which points they touch, and therefore in cost and in stability.

You usually encounter these in the **viscous terms of Navier–Stokes** on a non-orthogonal mesh.

# Differencing a whole PDE

For illustration purposes, we will consider the 1D, unsteady heat conduction:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

It is forward in time, central in space.

Every term is expanded about the same point $(n, j)$:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{(\Delta x)^2} \left( u_{i+1}^n - 2u_i^n + u_{i-1}^n \right)$$

This is the simple **explicit scheme**.

We have 1 unknown, $u$ at level $n + 1$, written in terms of known values at level $n$.

# Truncation error of the explicit scheme

$$\underbrace{\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2}}_{\text{PDE}} - \underbrace{\left[ \frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{\alpha}{(\Delta x)^2} \left( u_{i+1}^n - 2u_i^n + u_{i-1}^n \right) \right]}_{\text{FDE}} = \text{T.E.}$$

$$\text{T.E.} = - \left. \frac{\partial^2 u}{\partial t^2} \right|_i^n \frac{\Delta t}{2} + \alpha \left. \frac{\partial^4 u}{\partial x^4} \right|_i^n \frac{(\Delta x)^2}{12} + \cdots = O\left[ \Delta t, \, (\Delta x)^2 \right]$$

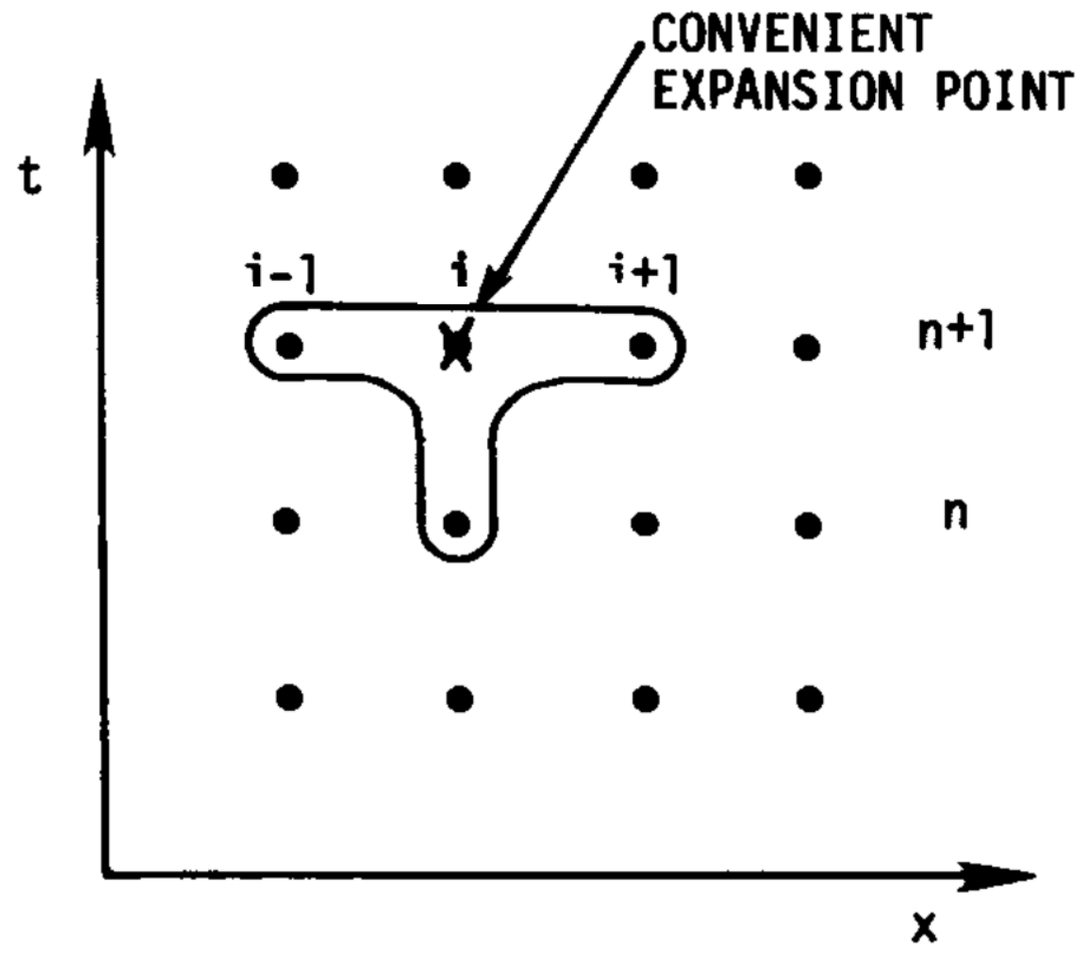
This is first order in time, second order in space.

Refining $\Delta x$ alone stops helping once the $\Delta t$ term dominates.

# Implicit scheme

Evaluate the space derivative at level $n+1$ instead, and three unknowns appear in a single equation:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{(\Delta x)^2}\left(u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}\right), \qquad \text{T.E.} = O\left[\Delta t, (\Delta x)^2\right]$$

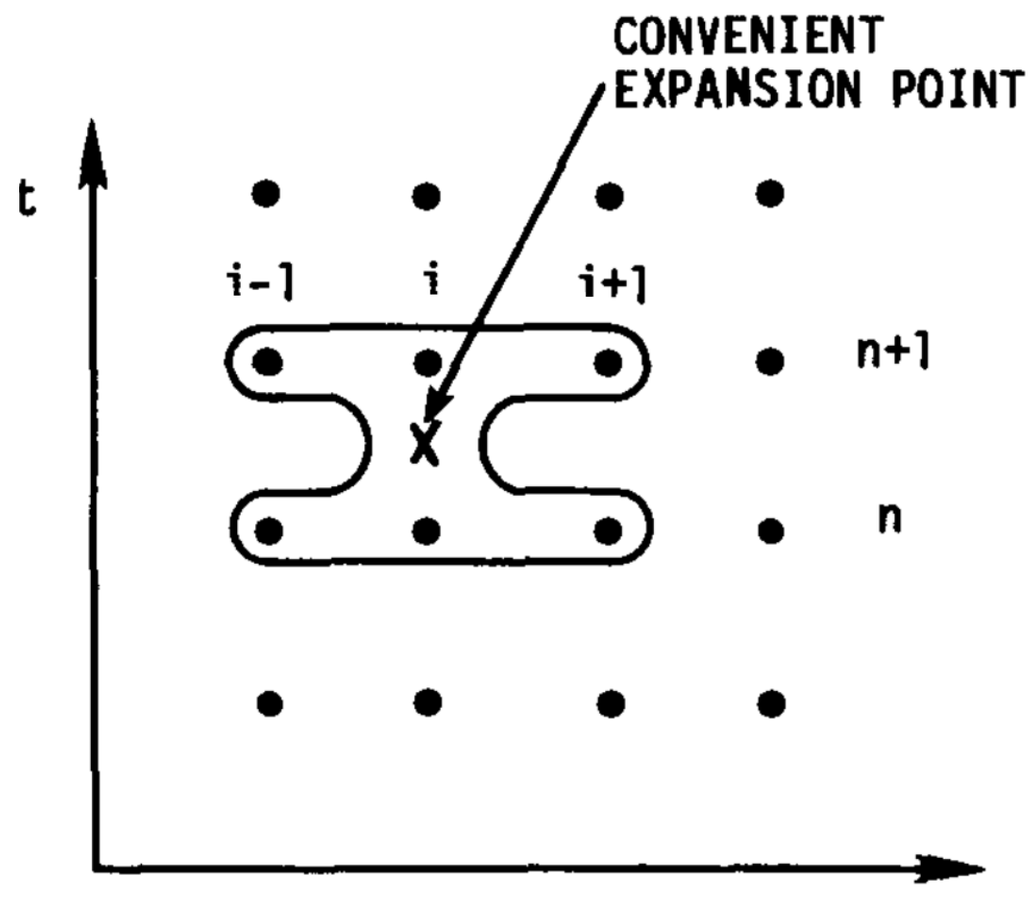



*Fully implicit form of the heat equation (Tannehill).*

# Crank–Nicolson scheme

Average the space derivative over the two-time levels, and the scheme becomes second order in time as well:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{2(\Delta x)^2}\left[u_{i+1}^{n+1} + u_{i+1}^n - 2\left(u_i^{n+1} + u_i^n\right) + u_{i-1}^{n+1} + u_{i-1}^n\right]$$




*Crank–Nicolson form (Tannehill).*

# Round-off and discretisation errors

**Round-off error:**

From finite-precision arithmetic (machine precision, machine epsilon error).
It often grows with the number of grid points.

**Discretisation error:**

Exact solution of the PDE minus exact solution of the FDE.
It is caused by the T.E. and by the boundary treatment.

What you actually obtain as the solution contains the sum of the two errors.

Refining the mesh reduces the discretisation error and increases the round-off error.

There is an optimum, and it is not $\Delta x \rightarrow 0$.

Analysing it and finding it is done in a **convergence study**.

# Consistency

A finite difference representation is consistent if its truncation error vanishes as the mesh is refined:

$$\lim_{\Delta x,\, \Delta t \to 0} (\mathrm{PDE} - \mathrm{FDE}) = \lim_{\Delta x,\, \Delta t \to 0} (\mathrm{T.E.}) = 0$$

**Nearly every scheme built by Taylor expansion is consistent, by construction.**
The exceptions are instructive.

**Exception: DuFort–Frankel scheme.**

Its T.E. carries $\alpha(\Delta t/\Delta x)^2$, consistent only if $\Delta t/\Delta x \to 0$: at a fixed ratio it solves a wave equation, not the heat equation.

**Important question**:
in the limit, am I solving the equation I meant to solve?

# Stability

A scheme is stable if errors from any source — round-off, truncation, a mistake in the initial data — are not amplified as the march proceeds.

### Applies to:

Marching problems only. Parabolic and hyperbolic equations.

### Cost:

Far more of our time than consistency.

It is subtle and it must be established; it cannot be assumed.

**Simple explicit:** stable only if $r = \alpha\Delta t/(\Delta x)^2 \leq ½$.

**Fully implicit:** unconditionally stable.

# The explicit stability limit

$$r \equiv \frac{\alpha\,\Delta t}{(\Delta x)^2} \leq \frac{1}{2} \qquad \Longleftrightarrow \qquad \Delta t \leq \frac{(\Delta x)^2}{2\alpha}$$

The restriction is quadratic in $\Delta x$.

Notice that if we halve the spatial step, then we must quarter the time step, which means four times the work per step, eight times in total.

This is the reason why implicit methods exist, and why diffusion is almost always treated implicitly in production codes.

# Better truncation error does not mean stability

Central in time as well as in space — formally second order in both:

$$\frac{u_i^{n+1} - u_i^{n-1}}{2\,\Delta t} = \frac{\alpha}{(\Delta x)^2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)$$

$$\text{T.E.} = O\left[(\Delta t)^2, (\Delta x)^2\right]$$

It is **unconditionally unstable**.

There is no $\Delta t$, however small, for which it can be used.

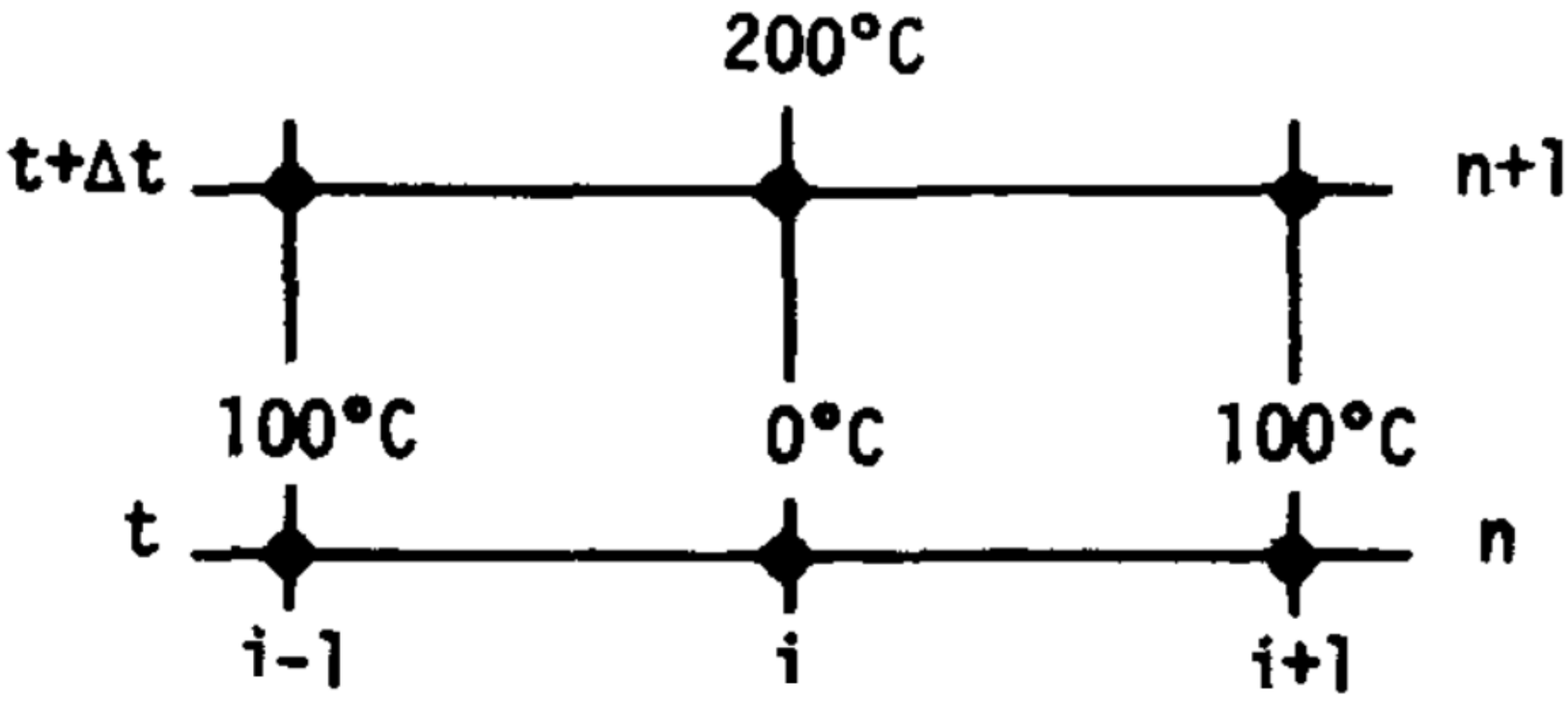
Accuracy and stability are independent properties.

A scheme can be more accurate and still worthless.

# Instability as physical implausibility

Let’s rearrange the explicit scheme so the unknown stands alone:

$$u_i^{n+1} = r\left(u_{i+1}^n + u_{i-1}^n\right) + (1-2r)\,u_i^n\,, \qquad r = \frac{\alpha\,\Delta t}{(\Delta x)^2}$$




*Physical implausibility resulting from r = 1 (Tannehill).*

$$r = 1: \qquad u_i^{n+1} = 1(100+100) + (1-2)(0) = 200\,^\circ\mathrm{C}$$

# Lax's equivalence theorem

Given a properly posed initial value problem and a consistent finite-difference approximation to it:

> Stability is the necessary and sufficient condition for convergence.

**Convergence:**
The solution of the FDE approaches the solution of the PDE as the mesh is refined. That is what we actually want.

**Theorem proved for:**
linear initial-value problems.

**Not proved for: :**
nonlinear problems — and we work almost exclusively with non-linear ones.

Everyone proceeds as though it held, so be careful here.

It is the justification for checking consistency and stability rather than convergence.

# Equilibrium problems: elliptical PDEs

Elliptic problems have no marching direction, so there is no sequence in which an error can grow.

Stability, in the strict sense, does not apply to elliptical problems.

**Consistency:**
It still applies, to every class of PDE equation.

**Iteration convergence:**
The iterative solver must converge, for arbitrarily fine meshes:

$$\left| u_{i,j}^{m+1} - u_{i,j}^{m} \right| < \varepsilon$$ at every grid point, iteration $m$.

Iteration convergence and truncation convergence are different.

Both are needed when solving PDEs.

# Conservation form

A PDE is in **conservation form when every spatial derivative can be identified as a divergence**.

Let's consider the **Continuity** equation, for example:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} = 0$$

$$\iff \qquad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \, \mathbf{v}) = 0$$

Expanding the products gives an equally valid, equally exact, **non-conservative form**:

$$\frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + \rho\frac{\partial u}{\partial x} + v\frac{\partial \rho}{\partial y} + \rho\frac{\partial v}{\partial y} + w\frac{\partial \rho}{\partial z} + \rho\frac{\partial w}{\partial z} = 0$$

# Why does the form matter?

The same choice for 1D conduction with variable properties:

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) \qquad \text{versus} \qquad \rho c \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} + \frac{\partial k}{\partial x}\frac{\partial T}{\partial x}$$

**Conservative form:**

the RHS is minus the divergence of the heat flux. Valid even where $k$ jumps.

**Non-conservative:**

this requires $\partial k/\partial x$ to exist. At a material interface it does not.

**At discontinuous coefficients, shocks, contact surfaces:**

the divergence form survives, while the expanded form does not.

This is the most important reason why CFD codes are written in conservation form.

# The conservative property of a scheme

The **conservative property** has to do not with the form of the PDE, but with whether the scheme enforces conservation exactly over any region.

$$\nabla \cdot (\rho \mathbf{v}) = 0 \implies \iiint_V \nabla \cdot (\rho \mathbf{v})\, dV = \oiint_S \rho\, \mathbf{v} \cdot \mathbf{n}\, dS = 0$$

For the discrete test, we write the scheme as a difference of face fluxes, multiplying by $\Delta x$ and sum over all $N$ cells:

$$\frac{f_{i+1/2} - f_{i-1/2}}{\Delta x} = 0\,, \qquad f_{i+1/2} = f(u_i,\, u_{i+1})$$

$$\sum_{i=1}^{N} \left(f_{i+1/2} - f_{i-1/2}\right) = \left(f_{3/2} - f_{1/2}\right) + \left(f_{5/2} - f_{3/2}\right) + \cdots + \left(f_{N+1/2} - f_{N-1/2}\right)$$

$$= f_{N+1/2} - f_{1/2} \quad \iff \quad \oiint_S \rho\, \mathbf{v} \cdot \mathbf{n}\, dS = 0$$

Everything in the interior disappears and only the two end faces survive (telescoping property). It is the divergence theorem on any mesh and for any number of cells.

# How much does it matter?

**Crucial when:**

a discontinuity is present. The integral form fixes the speed at which it travels:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \qquad \Longrightarrow \qquad \frac{d}{dt}\int_{x_L}^{x_R} u\, dx + f(u_R) - f(u_L) = 0$$

$$\Longrightarrow \qquad \boxed{S = \frac{f(u_R) - f(u_L)}{u_R - u_L}} \qquad \textbf{Rankine-Hugoniot}$$

A scheme that does not enforce the integral statement discretely has no reason to reproduce $S$, and refining the mesh would not fix this.

**Important note:**

A scheme can conserve perfectly and still propagate information in the wrong direction.

Any consistent scheme is adequate on a fine enough mesh — for smooth solutions. The difficulty of compressible CFD is precisely evaluating: "smooth" and "fine enough".

# Summary

### Truncation error:

T.E = PDE − FDE. It tells us the rate of improvement, not the size of the error.

### Consistency:

T.E. → 0 under refinement. Cheap to check, and DuFort–Frankel is an exception.

### Stability:

Errors must not grow. Independent of accuracy; $r \leq ½$ for the simple explicit scheme.

### Convergence:

This is what we always want to check. Lax theorem is applicable to linear problems only.

### Conservation:

Divergence form for the PDE with telescoping fluxes for the scheme. It is indispensable with shocks (as we will study later).